ONE-PAGE BRIEF

# Post-quantum cryptography review: fireblocks-mpc-lib @ 00ae08b7f1bd

Client: Fireblocks — public repository (passive review) · Source: github.com/fireblocks/mpc-lib, commit `00ae08b7f1bd` · Reviewed: 2026-11-06 14:44 UTC · Risk profile: Crypto exchange, custodian or wallet

POST-QUANTUM MATURITY
**Level 0 of 4**
no post-quantum cryptography

FINDINGS
**0** Critical **0** High **7** Medium
**0** Low **0** Info

QUANTUM TIMELINE (MOSCA: X + Y − Z)
Signatures: 10 + 5 − 8 = 7 → **too late**

**Next step:** Inventory where classical public-key cryptography protects long-lived data and keys, and choose the first migration: usually a hybrid TLS key exchange and wrapped keys.

## WHAT MATTERS MOST

| | | |
|---|---|---|
| Medium | QSI-2026-19319 | **Quantum-vulnerable Paillier: threshold signing and MPC protocols (30 file(s))** confirmed by the analyst<br>What to do: Track post-quantum threshold signatures (NIST IR 8214C) and keep the protocol layer ready to swap the signature scheme; record which keys... |
| Medium | QSI-2026-19320 | **Quantum-vulnerable Pedersen commitment: threshold signing and MPC protocols (8 file(s))**<br>confirmed by the analyst<br>What to do: Track post-quantum threshold signatures (NIST IR 8214C) and keep the protocol layer ready to swap the signature scheme; record which keys... |
| Medium | QSI-2026-19321 | **Quantum-vulnerable Threshold ECDSA: threshold signing and MPC protocols (7 file(s))**<br>confirmed by the analyst<br>What to do: Track post-quantum threshold signatures (NIST IR 8214C) and keep the protocol layer ready to swap the signature scheme; record which keys... |
| Medium | QSI-2026-19322 | **Quantum-vulnerable FROST: threshold signing and MPC protocols (7 file(s))** confirmed by the analyst<br>What to do: Track post-quantum threshold signatures (NIST IR 8214C) and keep the protocol layer ready to swap the signature scheme; record which keys... |

3 more in the full report.

## WHAT THE QUANTUM-VULNERABLE CRYPTOGRAPHY PROTECTS

| Role | Algorithms | What to do |
|---|---|---|
| Medium threshold signing and MPC protocols | Paillier, Pedersen commitment, FROST, Threshold ECDSA ... | Track post-quantum threshold signatures (NIST IR 8214C) and keep the protocol layer ready to swap the... |
| Medium role not determined | Ed25519 | Classify each usage, record data lifetimes, and plan migration to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205)... |

Analyst review: 7 confirmed, 0 not applicable, 0 dismissed as false positives. (QSI review, 2026-10-28)

Static review of source code and manifests; files analysed: 123. No requests were sent to the owner's systems.

Full report, content SHA-256: `e6f72f8f1970037a386b9f287905f53dd85d0a6429cd8202df276221dd534f3f`

Full report signature (report.sig.json): Ed25519 key, SHA-256 fingerprint `5b3cf9c9bd0fa54447e2b5dfa7a6bab569bd0c666fa46949599e5c348d17083e`. bundle.sig.json lists the SHA-256 of every file in this bundle and is signed too; verify-bundle checks it offline.

Preliminary findings need confirmation in the code; the full report gives the evidence, the limits of each check and how to re-test.